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Electrical and Computer Engineering

Part 1: Analog Active Filter

EEL3923C Design 1

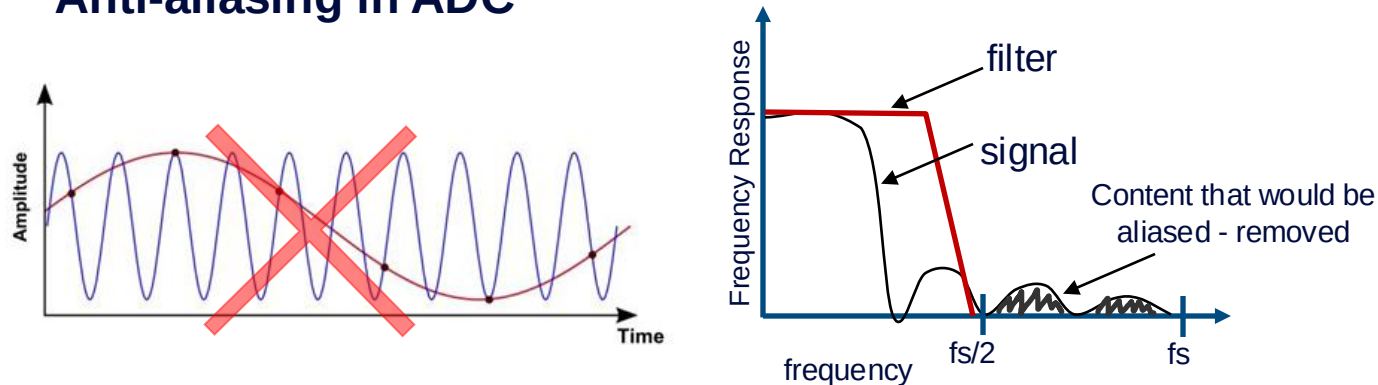
Erin Patrick, Michael Stapleton

POWERING THE NEW ENGINEER TO TRANSFORM THE FUTURE

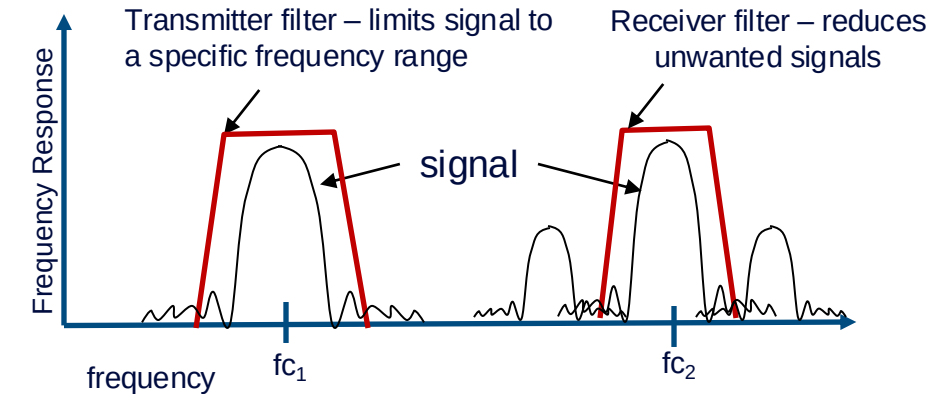
Analog Filters in Use

Filters can be used to separate signals, passing those of interest and attenuating the unwanted frequencies.

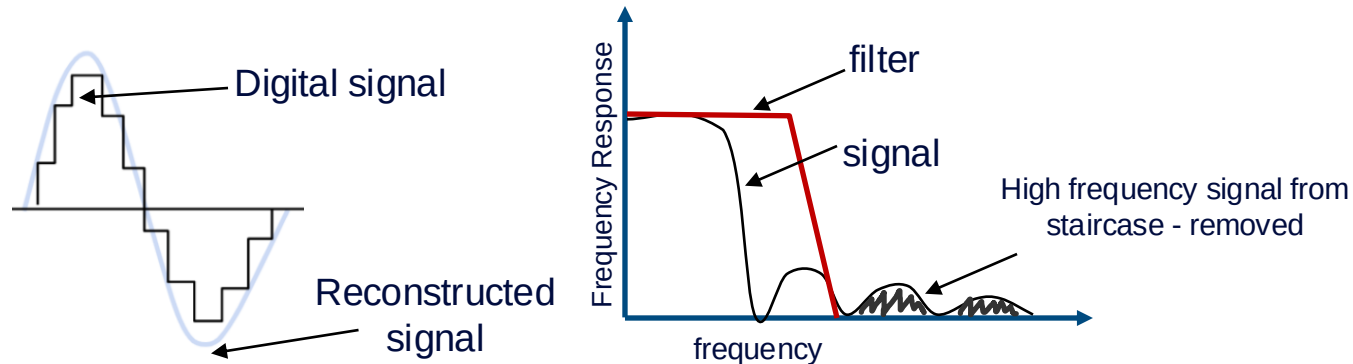
Anti-aliasing in ADC



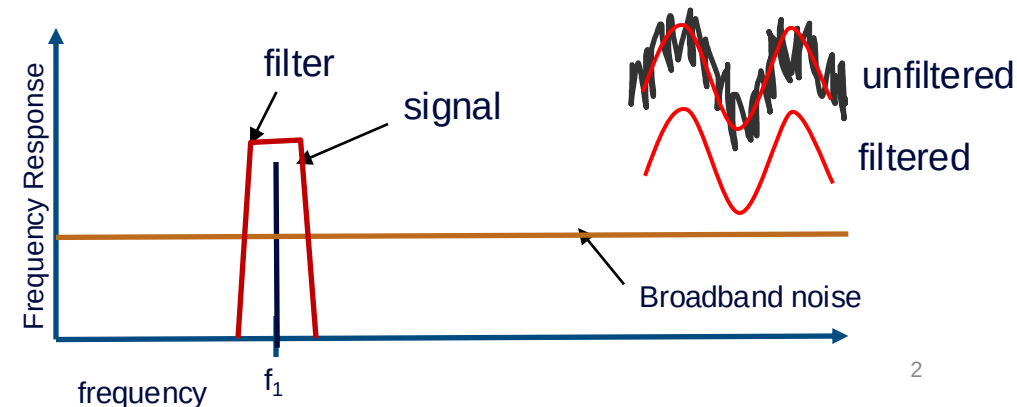
Signal specificity in radio transmitters and receivers



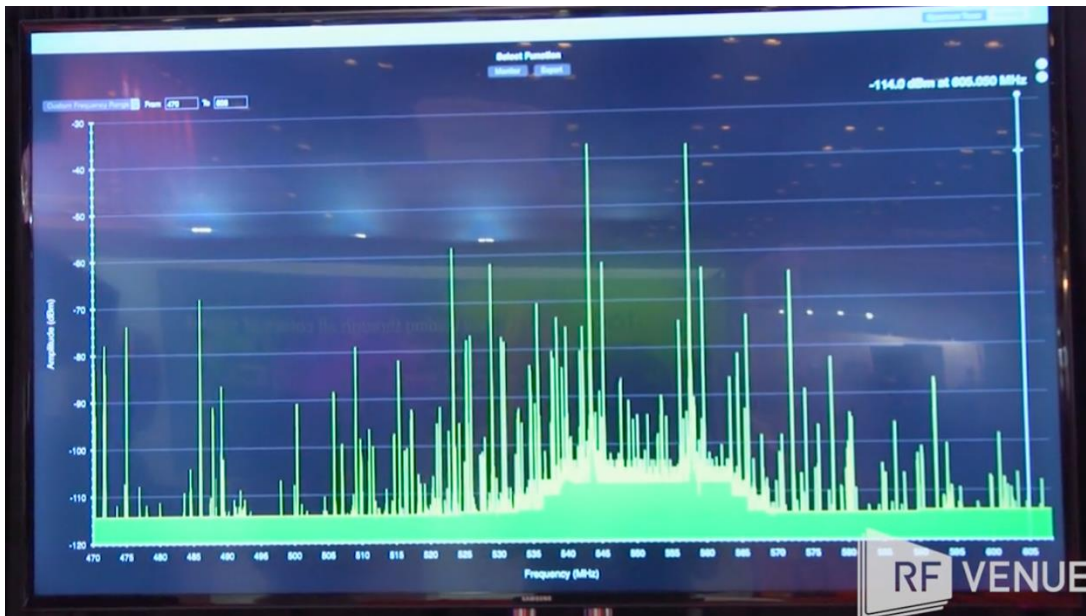
Smoothing (reconstruction) in DAC



Noise cancellation in instrumentation



Example: Use of bandpass filter in RF receiver for wireless mics



Unfiltered

A. Frequency Spectrum of all MHz signals picked up by custom antenna in a conference hall.



Filtered

B. Frequency Spectrum of all MHz signals after filtering by a bandpass filter in a conference hall. Noise has been reduced. Mics will not saturate.

Source: <https://www.rfvenue.com/blog/bandpass-filters-wireless-microphones>

Ideal Filter Types

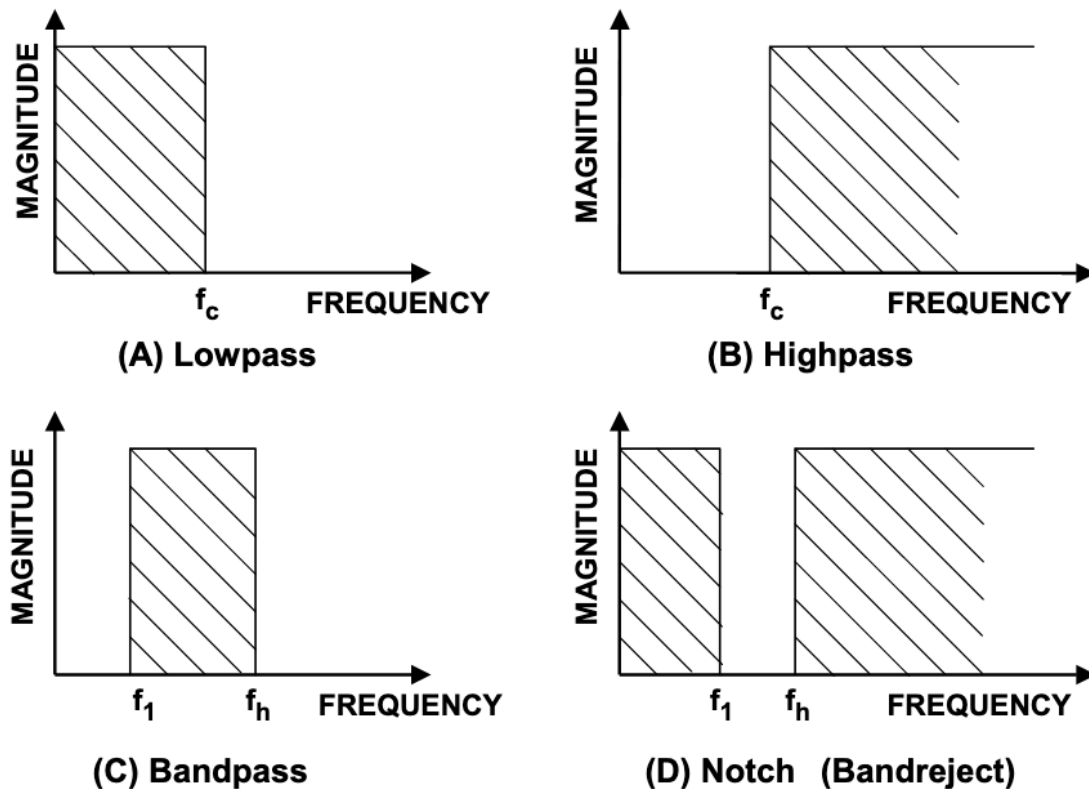


Figure 8.1: Idealized Filter Responses

Ex. Nonideal Lowpass Filter

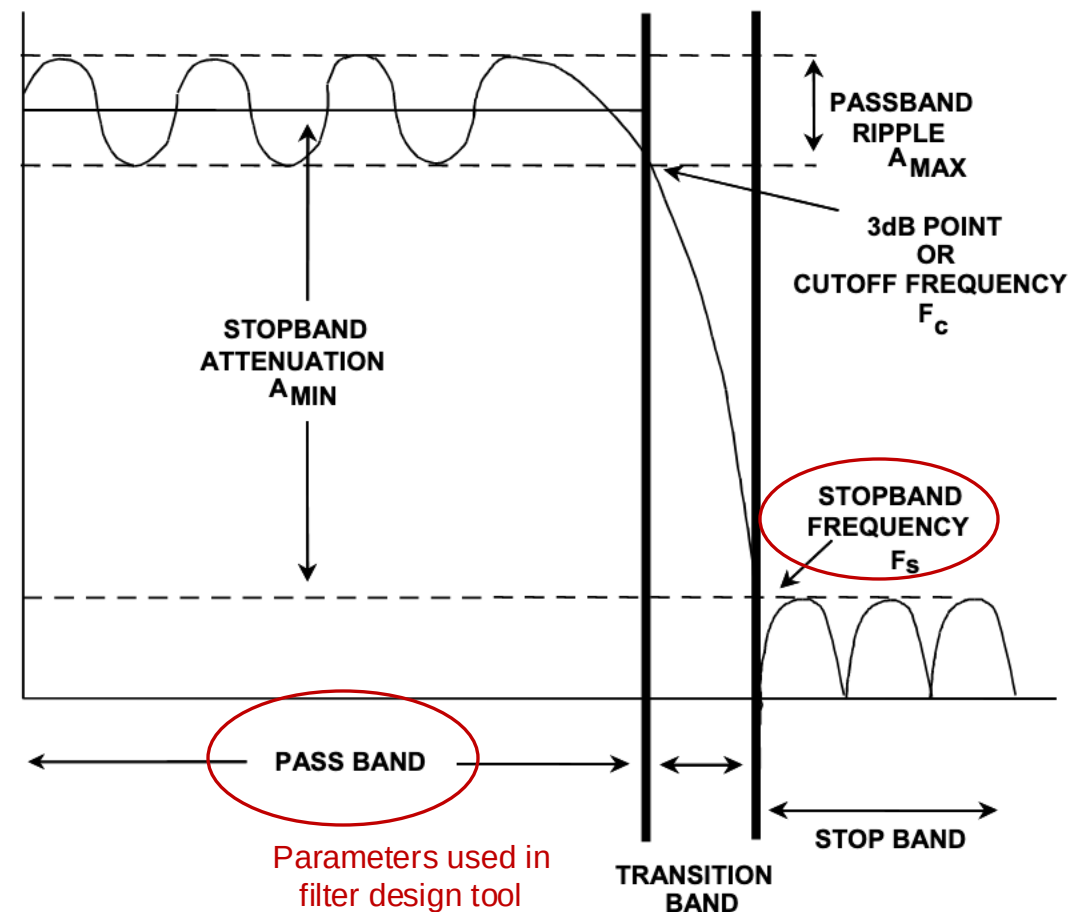
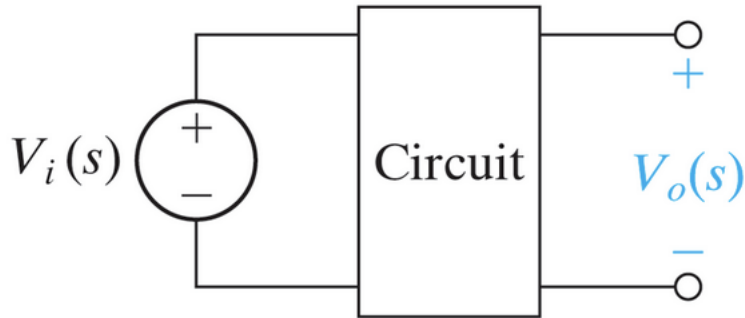


Figure 8.2: Key Filter Parameters

Review - Defining the Frequency Response of a Filter



Transfer Function $H(j\omega)$, s – relationship of the output over the input, in the frequency domain

$$H(j\omega) = \frac{V_o(j\omega)}{V_i(j\omega)}$$

ω is angular frequency
 $\omega = 2\pi f$ [rad/s]
 j is the imaginary number (i)

Using Laplace transform:
 $j\omega = s$

$$H(s) = \frac{V_o(s)}{V_i(s)}$$

Magnitude

Phase

$$|H(j\omega)| = \left| \frac{V_o(j\omega)}{V_i(j\omega)} \right|$$

$$\theta(j\omega) = \tan^{-1} \left[\frac{\text{Im}(H(j\omega))}{\text{Re}(H(j\omega))} \right]$$

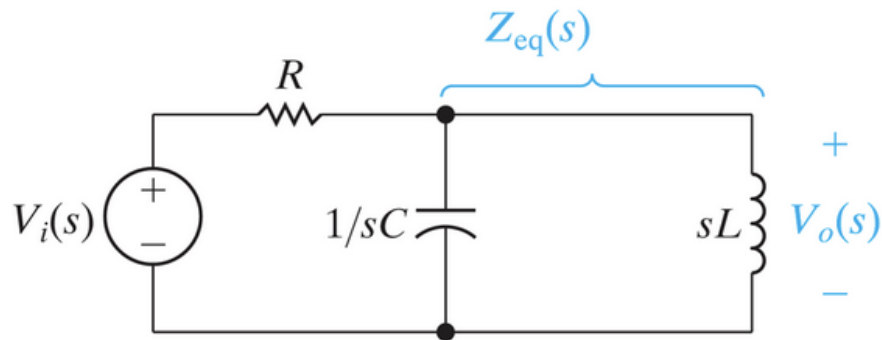
Magnitude of a complex number:

$$|(a + jb)| = \sqrt{a^2 + b^2}$$

* These are the quantities plotted on a Bode plot

Review: Passive Realization of RLC bandpass filter

Parallel RC combination



$$H(s) = \frac{V_o(s)}{V_i(s)} = \frac{Z_{eq}}{R + Z_{eq}} = \frac{\frac{s}{RC}}{s^2 + \frac{s}{RC} + \frac{1}{LC}}$$

$$Z_{eq} = \frac{sL(\frac{1}{sC})}{sL + \frac{1}{sC}} = \frac{sL}{LCs^2 + 1}$$

Characteristic form for 2nd order filter

- One zero (s in numerator)
- Two poles (s² in denom.)

$$|H(j\omega)| = \frac{\frac{\omega}{RC}}{\sqrt{\left(\frac{1}{LC} - \omega^2\right)^2 + \left(\frac{\omega}{RC}\right)^2}}$$

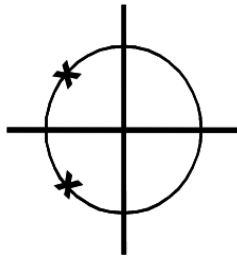
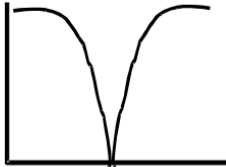
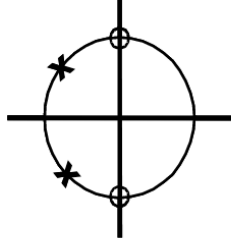
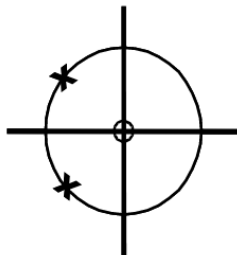
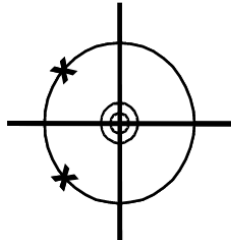
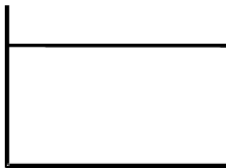
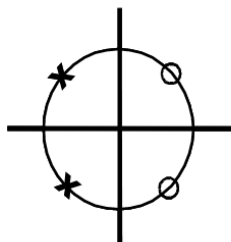
$$\omega_o = \sqrt{\omega_{c1} \cdot \omega_{c2}}$$

$$\omega_{c1} = -\frac{1}{2RC} + \sqrt{\left(\frac{1}{2RC}\right)^2 + \frac{1}{LC}},$$

$$\omega_{c2} = \frac{1}{2RC} + \sqrt{\left(\frac{1}{2RC}\right)^2 + \frac{1}{LC}}.$$

$$\omega_o = \sqrt{\frac{1}{LC}}.$$

Transfer Functions of Multiple 2nd Order Filter Types (s-domain)

FILTER TYPE	MAGNITUDE	POLE LOCATION	TRANSFER EQUATION	NOTCH (BANDREJECT)			
LOWPASS			$\frac{\omega_0^2}{s^2 + \frac{\omega_0}{Q}s + \omega_0^2}$			$\frac{s^2 + \omega_z^2}{s^2 + \frac{\omega_0}{Q}s + \omega_0^2}$	
BANDPASS			$\frac{\frac{\omega_0}{Q}s}{s^2 + \frac{\omega_0}{Q}s + \omega_0^2}$			$\frac{s^2}{s^2 + \frac{\omega_0}{Q}s + \omega_0^2}$	
				ALLPASS			$\frac{s^2 - \frac{\omega_0}{Q}s + \omega_0^2}{s^2 + \frac{\omega_0}{Q}s + \omega_0^2}$

Source: <https://www.analog.com/media/en/training-seminars/design-handbooks/Basic-Linear-Design/Chapter8.pdf>

Generic Transfer Function

$$H(s) = \frac{a_M s^M + a_{M-1} s^{M-1} + \dots + a_0}{s^N + b_{N-1} s^{N-1} + \dots + b_0}$$

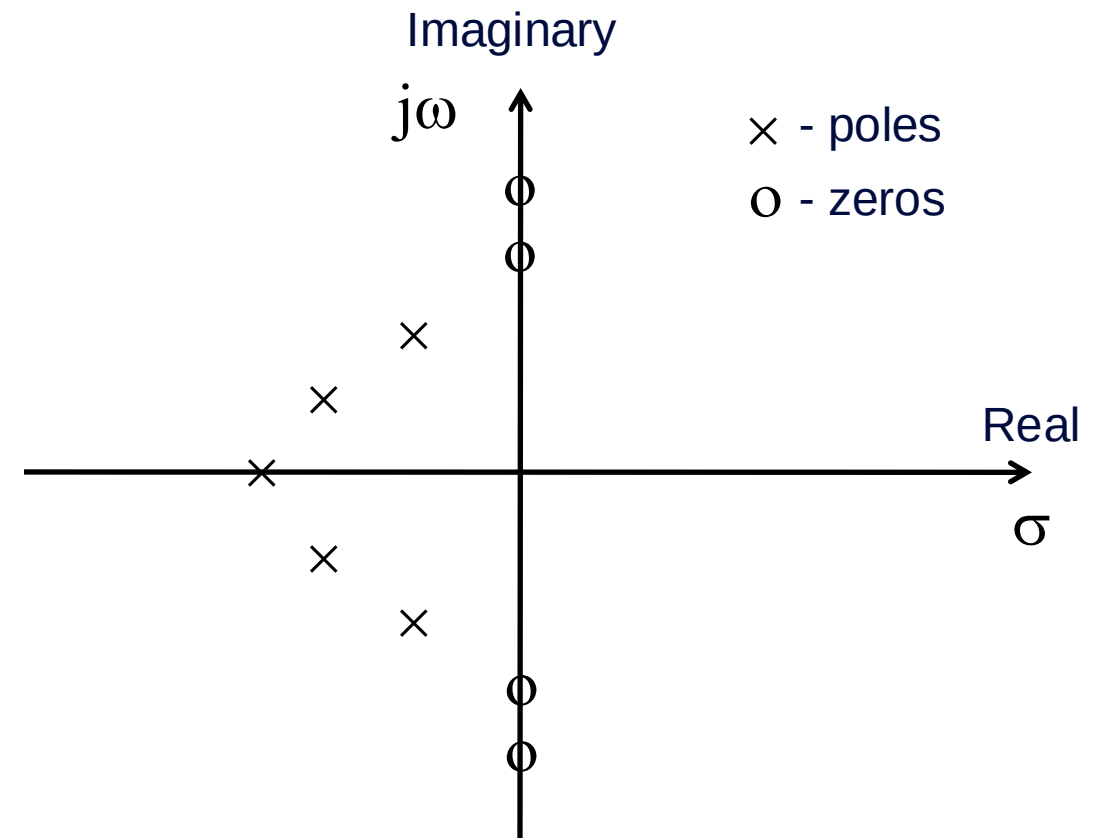
$$H(s) = \frac{a_M (s - z_1) \cdot (s - z_2) \cdot \dots \cdot (s - z_M)}{(s - p_1) \cdot (s - p_2) \cdot \dots \cdot (s - p_M)}$$

Complex numbers in even order filters

- Come in complex conjugate pairs
- Odd order filters include one real pole

Pole-Zero Pattern

- Poles and zeros plotted on complex s-plane
- All poles are complex conjugate except for odd order filter (one pole on real axis)



Butterworth (maximally flat magnitude)

This filter has the flattest possible pass-band magnitude response. Attenuation is -3 dB at the design cutoff frequency. Attenuation above the cutoff frequency is a moderately steep 20-dB per decade per pole. The pulse response of the Butterworth filter has moderate overshoot and ringing.

Chebyshev (equal ripple magnitude)

Note: Mr. Chebyshev's name is also transliterated Tschebychev, Tschebyscheff or Tchevysheff. This filter type has steeper attenuation above the cutoff frequency than Butterworth. This advantage comes at the penalty of amplitude variation (ripple) in the passband. Unlike Butterworth and Bessel responses, which have 3-dB attenuation at the cutoff frequency, Chebyshev cutoff frequency is defined as the frequency at which the response falls below the ripple band. For even-order filters, all ripple is above the 0-dB-gain dc response, so cutoff is at 0-dB (see Figure 1.) For odd-order filters, all ripple is below the 0-dB-gain dc response, so cutoff is at $-(\text{ripple})$ dB (see Figure 2.) For a given number of poles, a steeper cutoff can be achieved by allowing more pass-band ripple. The Chebyshev has even more ringing in its pulse response than the Butterworth.

Bessel (maximally flat time delay)

(Also called Thomson.) Due to its linear phase response, this filter has excellent pulse response (minimal overshoot and ringing). For a given number of poles, its magnitude response is not as flat, nor is its attenuation beyond the -3 -dB cutoff frequency as steep as the Butterworth. Although it takes a higher-order Bessel filter to give a magnitude response which approaches that of a given Butterworth filter, the pulse response fidelity of the Bessel filter may make the added complexity (because of additional filter sections) worthwhile.

Filter Types used in Industry

- The coefficients in each transfer function polynomial (i.e. where the poles are located) determine the filter response and thus type

Butterworth Lowpass Filter

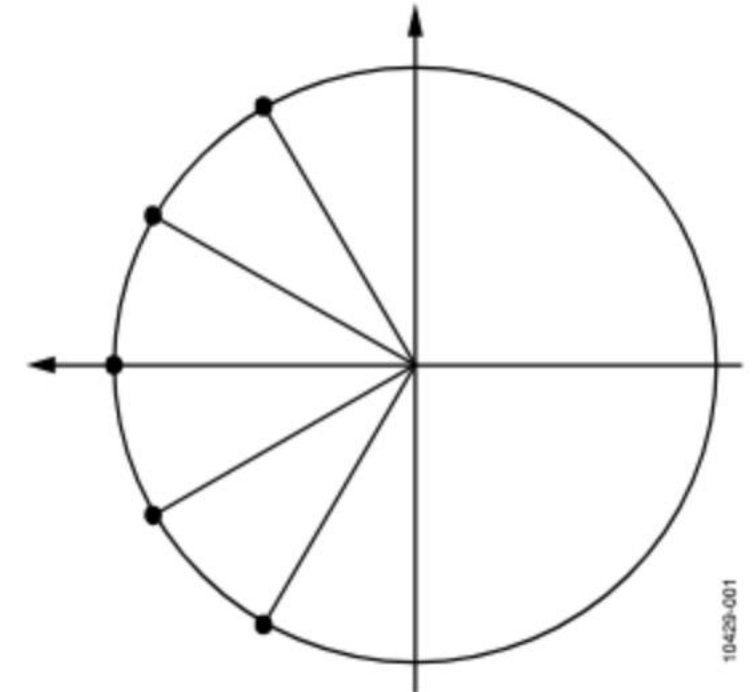
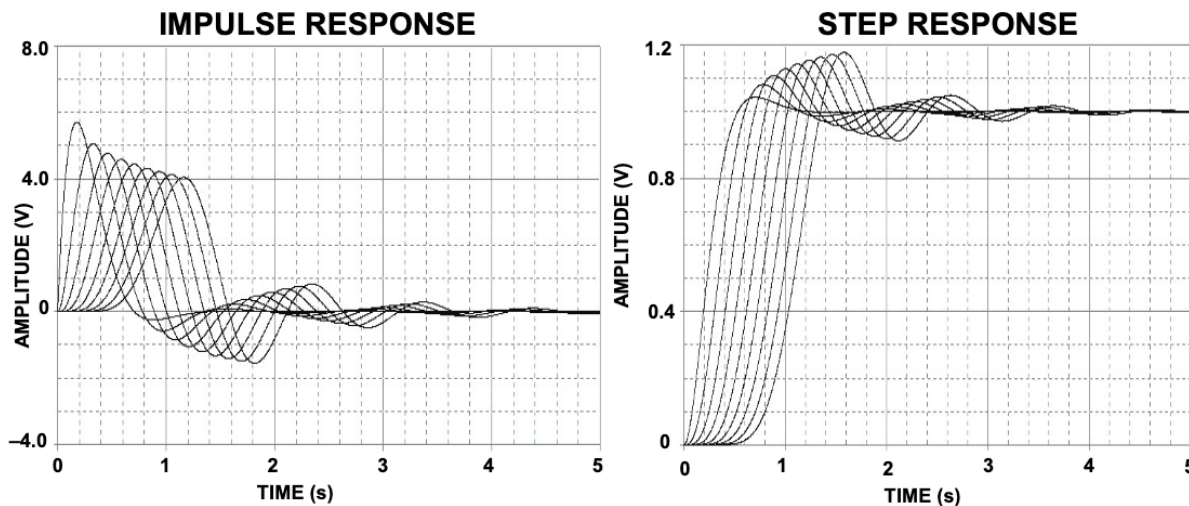
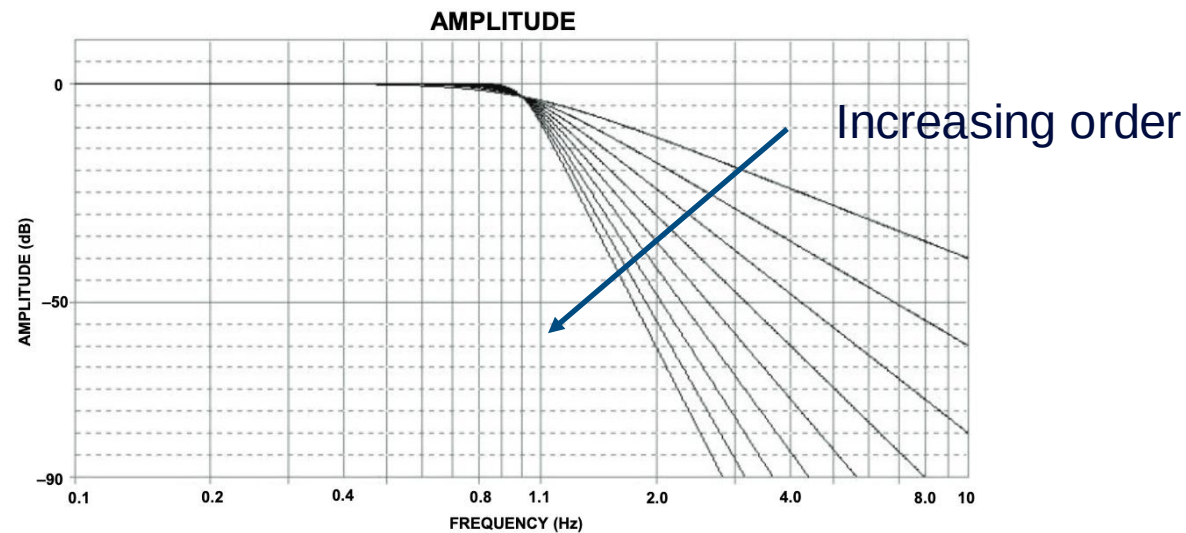
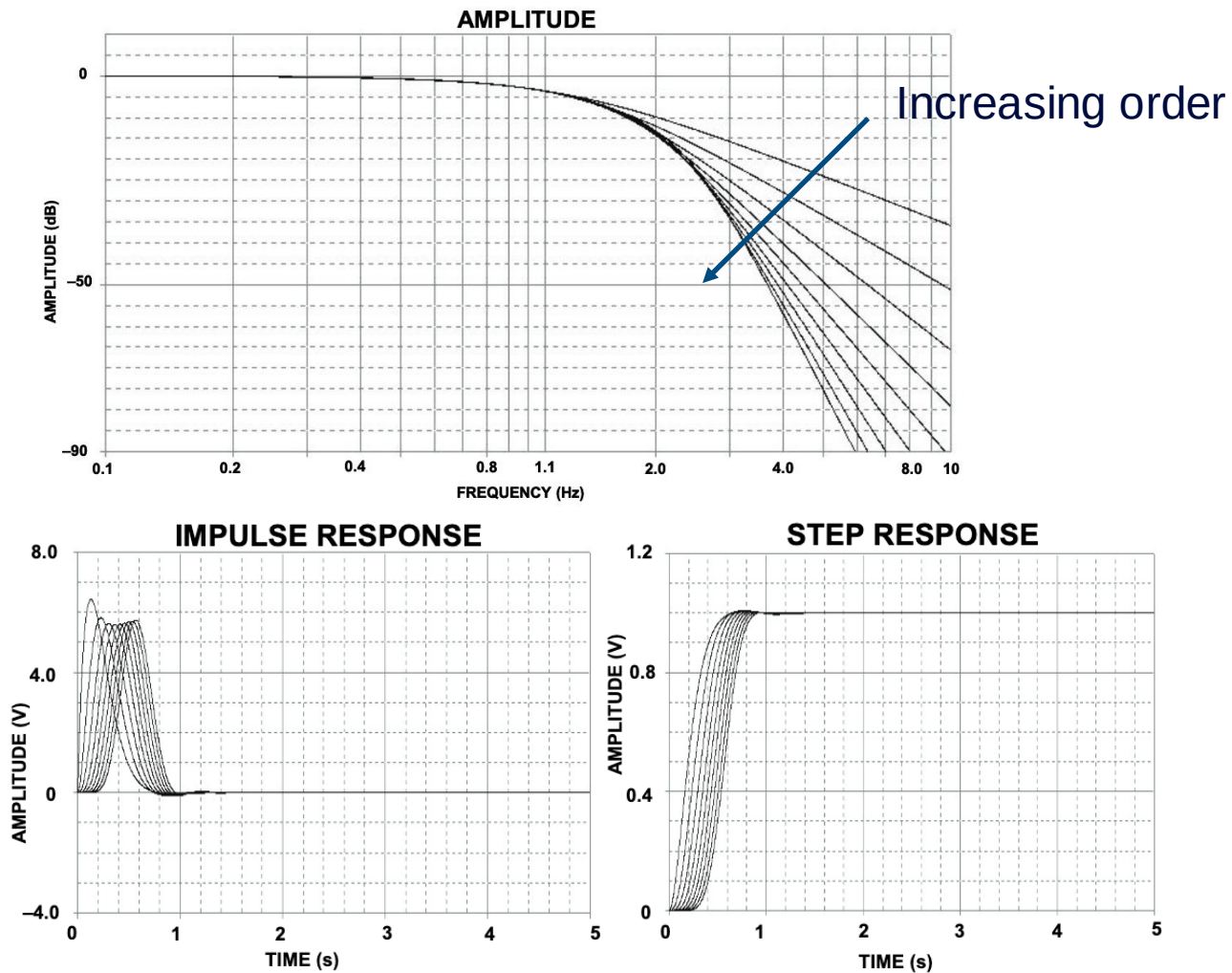


Figure 1. Butterworth Pole Location

Source: <https://www.analog.com/media/en/training-seminars/design-handbooks/Basic-Linear-Design/Chapter8.pdf>

Bessel Lowpass Filter



BESSEL POLE LOCATIONS

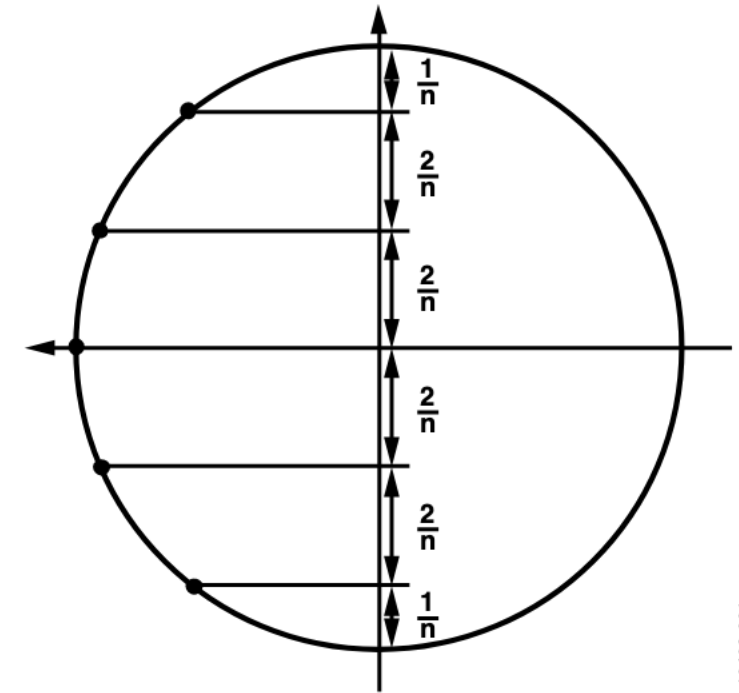


Figure 1. Bessel Pole Locations

10402-001

Source: <https://www.analog.com/media/en/training-seminars/design-handbooks/Basic-Linear-Design/Chapter8.pdf>

Chebyshev Lowpass Filter

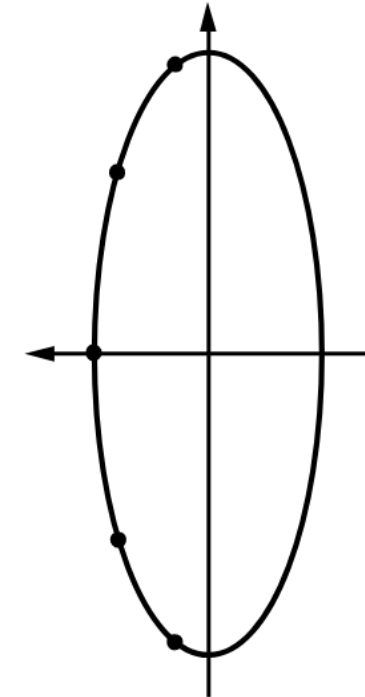
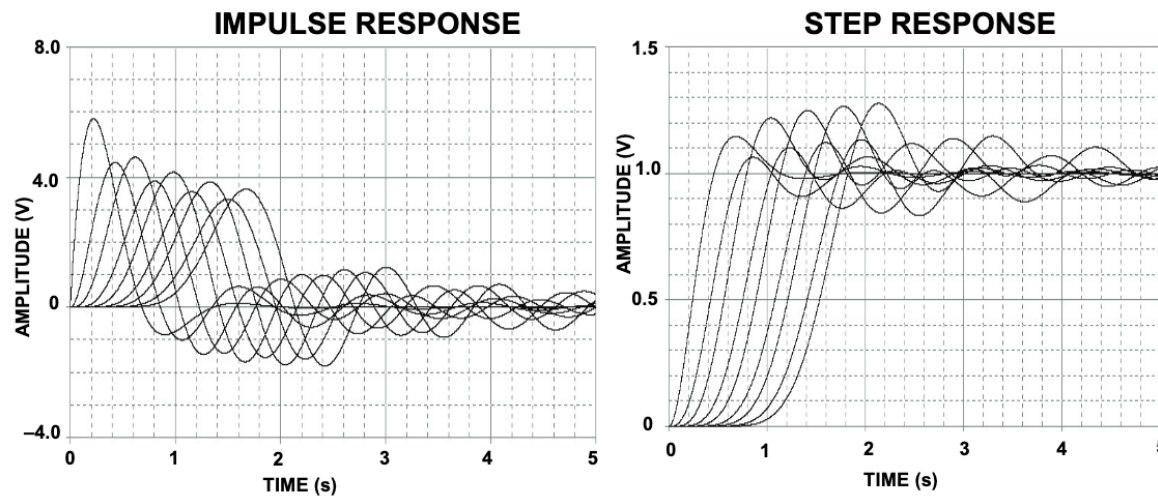
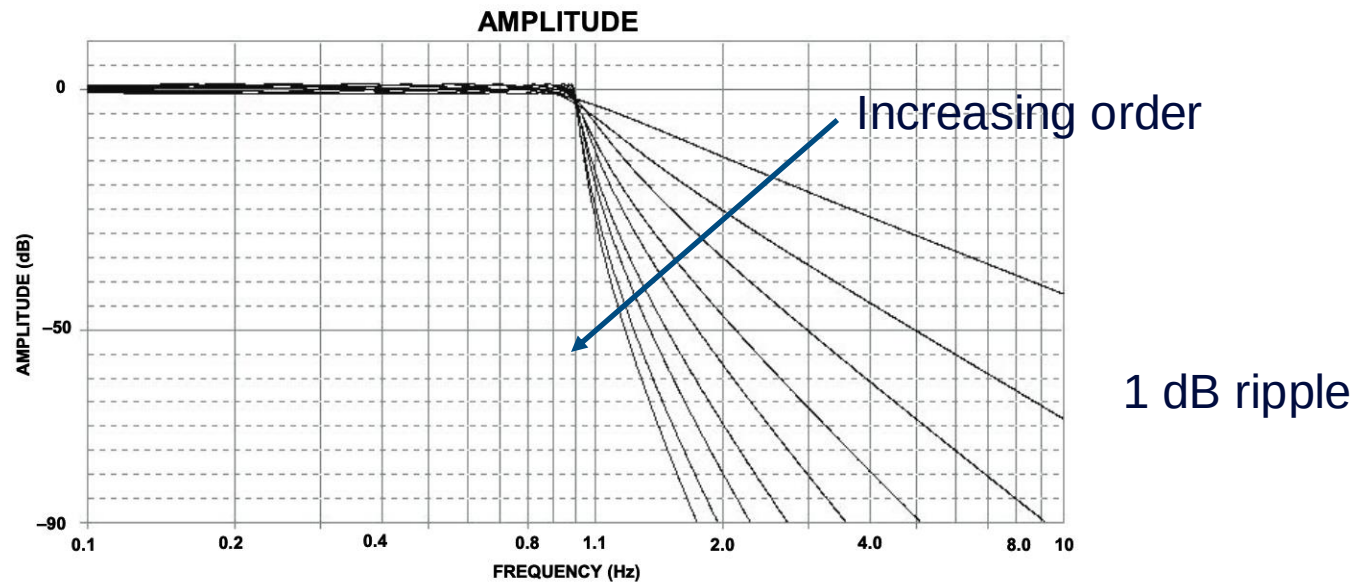
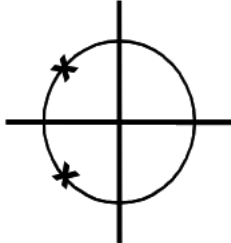


Figure 1. 1 dB Chebyshev Pole Location

General Transfer function for 2nd order LowPass Filter

FILTER TYPE	MAGNITUDE	POLE LOCATION	TRANSFER EQUATION
LOWPASS			$\frac{+H \omega_0^2}{s^2 + \alpha \omega_0 s + \omega_0^2}$

H = gain

$Q = 1/\alpha$ = quality factor (amount of “peaking” near cut-off frequency (resonant frequency), ω_0)

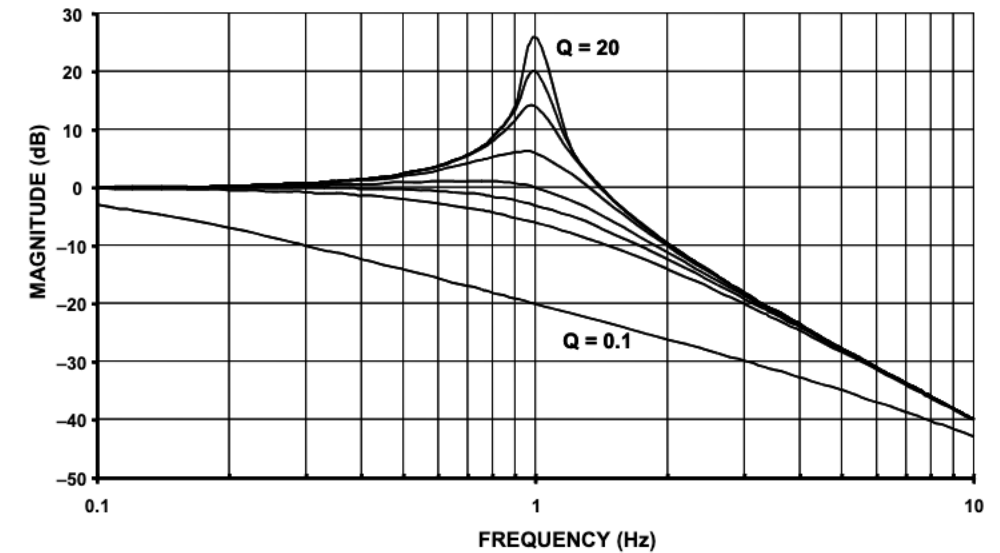


Figure 8.5: Low-Pass Filter Peaking vs. Q

Source (p. 12, 17): <https://www.analog.com/media/en/training-seminars/design-handbooks/Basic-Linear-Design/Chapter8.pdf>

Active Filter Topologies

- Active Filters made from op-amp circuits have many advantages
 - Loads do not influence the filter as they would in passive filters
 - Active filters do not need inductors
 - Saves space as inductors are bulky

*each topology has its advantages and disadvantages. See Analog Devices Chapter 8 link for more information

There are multiple circuit topologies that can make an active filter. Some include:

Sallen-key

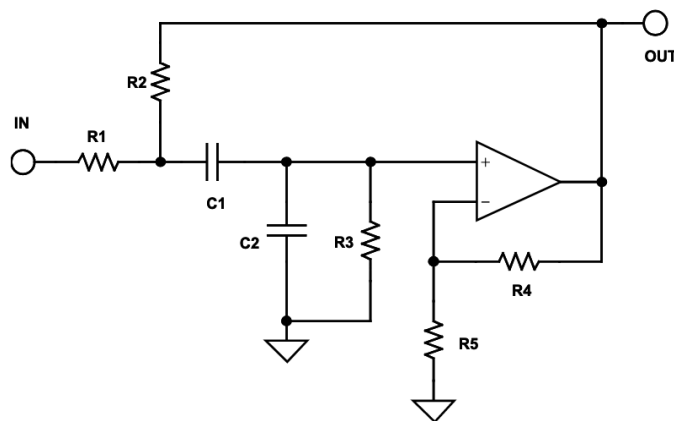


Figure 8.51: Sallen-Key Band-pass Filter

Multiple Feedback

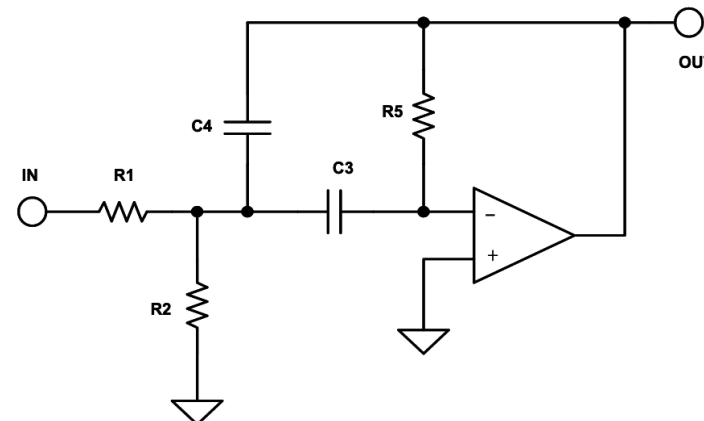


Figure 8.54: Multiple Feedback Band-Pass

State Variable

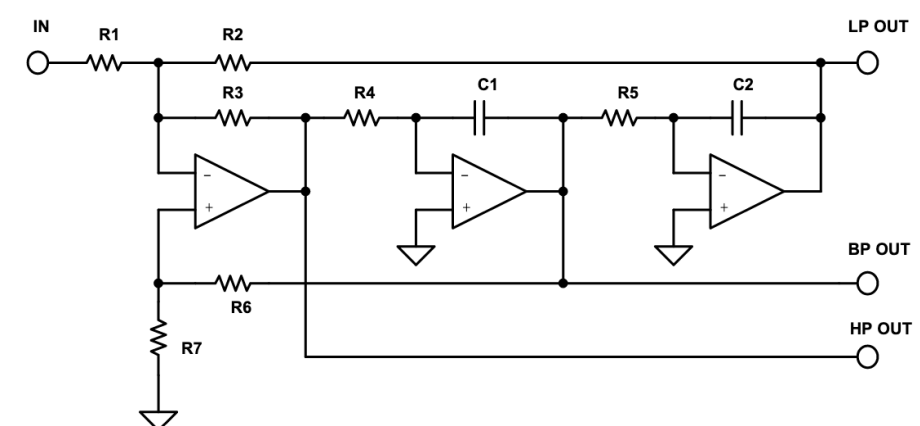
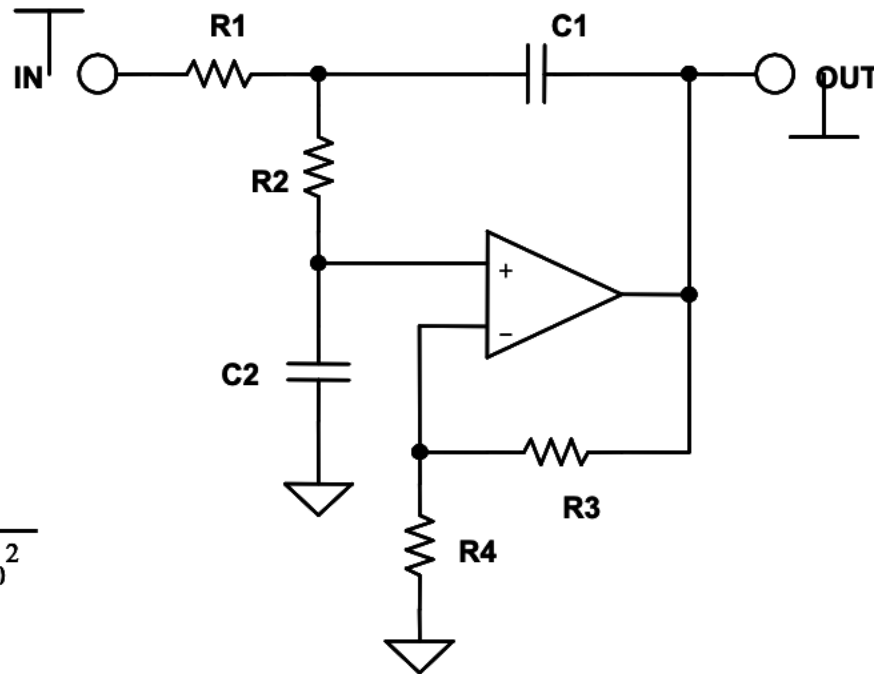


Figure 8.55: State Variable Filter

Filter Theory: 2nd order Lowpass, Sallen Key configuration

SALLEN-KEY LOWPASS



$$\frac{+H \omega_0^2}{s^2 + \alpha \omega_0 s + \omega_0^2}$$

$$\frac{V_O}{V_{IN}} = \frac{H \frac{1}{R_1 R_2 C_1 C_2}}{s^2 + s \left[\left(\frac{1}{R_1} + \frac{1}{R_2} \right) \frac{1}{C_1} + \frac{(1-H)}{R_2 C_2} \right] + \frac{1}{R_1 R_2 C_1 C_2}}$$

CHOOSE: C_1 R_3

THEN: $k = 2 \pi F_0 C_1$
 $m = \frac{\alpha^2}{4} + (H-1)$

$$R_4 = \frac{R_3}{(H-1)}$$

$$C_2 = m C_1$$

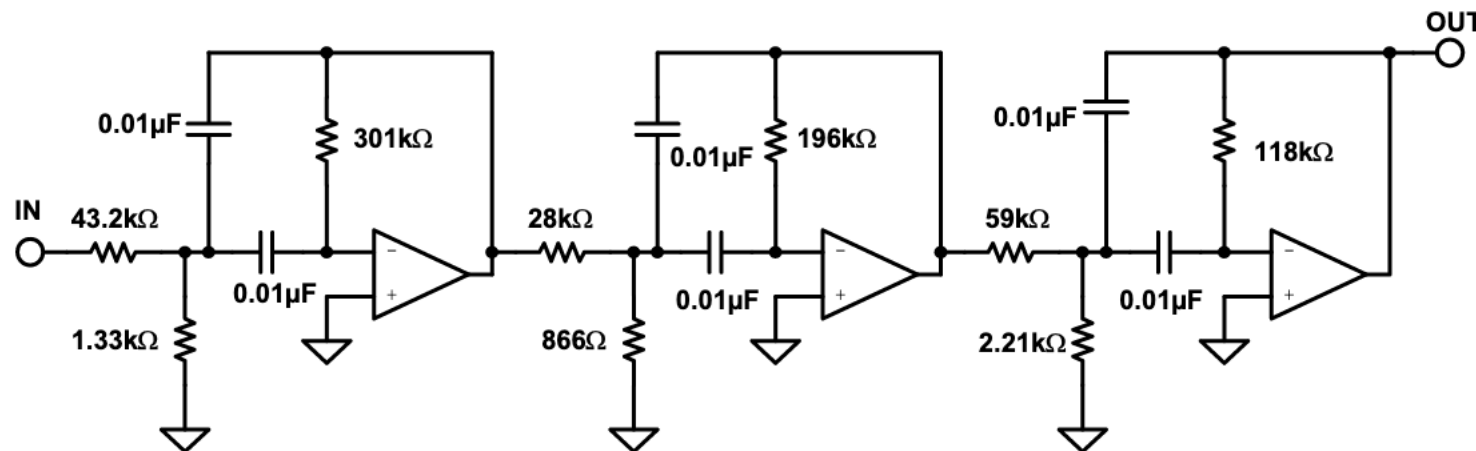
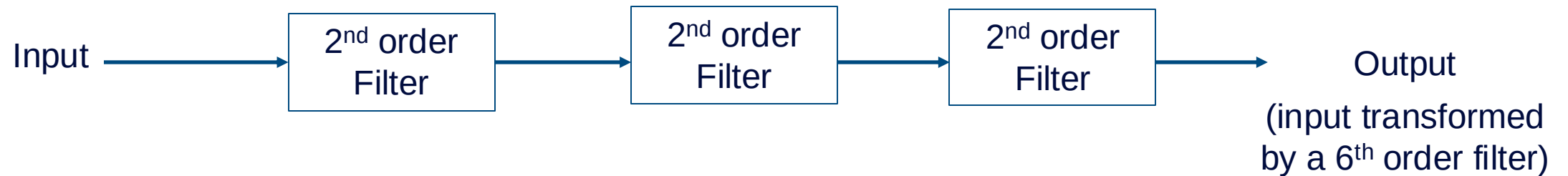
$$R_1 = \frac{2}{\alpha k}$$

$$R_2 = \frac{\alpha}{2mk}$$

Source (p. 93): <https://www.analog.com/media/en/training-seminars/design-handbooks/Basic-Linear-Design/Chapter8.pdf>

Higher Order Filter Realization

- Second order (circuits containing 2 capacitors) and first order filters can be cascaded to implement higher order filters



Stages using multiple feedback topology

Assignment (Analog Filter)

- Design an active, lowpass filter:
 - That will transform a square wave into a sine wave (frequency = 1000 Hz + last three digits of your UFid)
 - Design guides:
 - Python or Matlab simulation for signal processing (find Python starter script in Canvas)
 - Analog Devices Filter Tool or TI Filter tool (websites on Resources slide)
 - 0 dB gain
 - At least a 4th order filter (around -80 dB/dec slope) with 2 cascaded op-amps using Sallen Key or Multiple feedback configuration
- Simulate the filter in LT Spice
- Build the filter on the breadboard
 - Demonstrate that a square wave (frequency = 1000 Hz + last three digits of UFid) can be transformed into a sine wave with similar amplitude and frequency as square wave

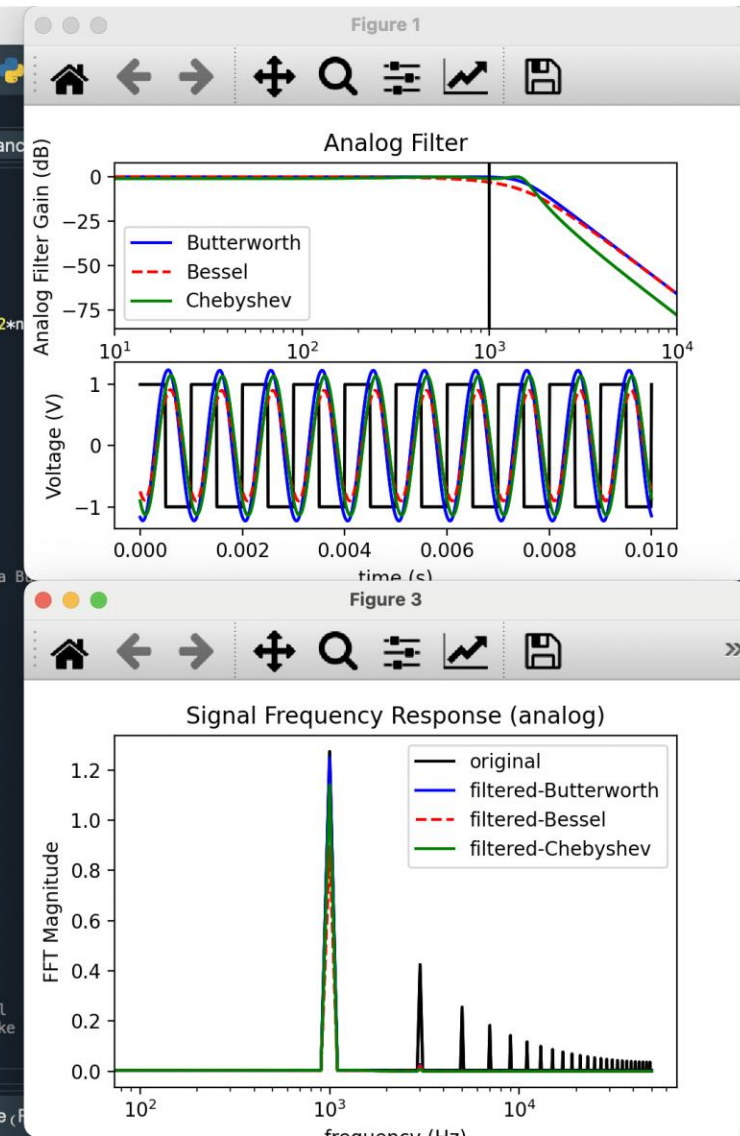
Python Script

FilterModule_lowpass.py

```

56 #-----Plot the filter response and time-domain filtered signal-----
57
58 #plot the frequency response of the analog filters
59
60 plt.figure(1)
61 plt.subplot(2,1,1)
62 plt.title('Analog Filter')
63 plt.semilogx(w/(2*np.pi), 20 * np.log10(abs(h)), "b", w2/(2*np.pi), 20 * np.log10(abs(h2)), "--r", w3/(2*np.pi), 20 * np.log10(abs(h3)), "g")
64 plt.xlabel('Frequency (Hz)')
65 plt.ylabel('Analog Filter Gain (dB)')
66 plt.margins(0, 0.1)
67 plt.grid(which='both', axis='both')
68 plt.axvline(fsq, color='k') # square wave frequency
69 plt.legend(['Butterworth', 'Bessel', 'Chebyshev'])
70
71 # Compute FFT of time-domain signals
72 fft_square = fft(square_sig) #complex value of the FFT spectrum
73 freq_pulse = fftfreq(len(t), d=dt) #frequency array, pass in length of signal and step size
74
75 # compute a transfer function of a filter type using the same frequencies as the FFT of the time signal
76 w, h = signal.freqs(b, a, worN=freq_pulse*2*np.pi) # change filter type as needed
77 w2, h2 = signal.freqs(b2, a2, worN=freq_pulse*2*np.pi) # change filter type as needed (i.e., a,b go to a B)
78 w3, h3 = signal.freqs(b3, a3, worN=freq_pulse*2*np.pi)
79
80 #Compute an inverseFFT of the filtered signal (multiply square FFT by filter transfer function )
81 filteredt1 = ifft(fft_square*h)
82 filteredt2 = ifft(fft_square*h2)
83 filteredt3 = ifft(fft_square*h3)
84
85 #plot square wave and filtered time domain wave
86
87 plt.subplot(2,1,2)
88
89 plt.plot(t, square_sig, '-k', t, (filteredt1), '-b', t, filteredt2, '--r', t, filteredt3, '-g')
90 #plt.legend(['original', 'filtered-Butterworth', 'filtered-Bessel', 'filtered-Chebyshev'])
91 plt.xlabel('time (s)')
92 plt.ylabel('Voltage (V)')
93
94 plt.show()
95
96 #-----Plot the frequency-domain filtered signal-----
97
98 # Multiply the FFT of the square wave by the transfer function of the filter to get the filtered signal
99 filtered1 = 2.0/N * np.abs(fft_square[0:int(len(freq_pulse)/2)]*(h[0:int(len(freq_pulse)/2)])) #only take
100 filtered2 = 2.0/N * np.abs(fft_square[0:int(len(freq_pulse)/2)]*(h2[0:int(len(freq_pulse)/2)]))
101 filtered3 = 2.0/N * np.abs(fft_square[0:int(len(freq_pulse)/2)]*(h3[0:int(len(freq_pulse)/2)]))
102

```

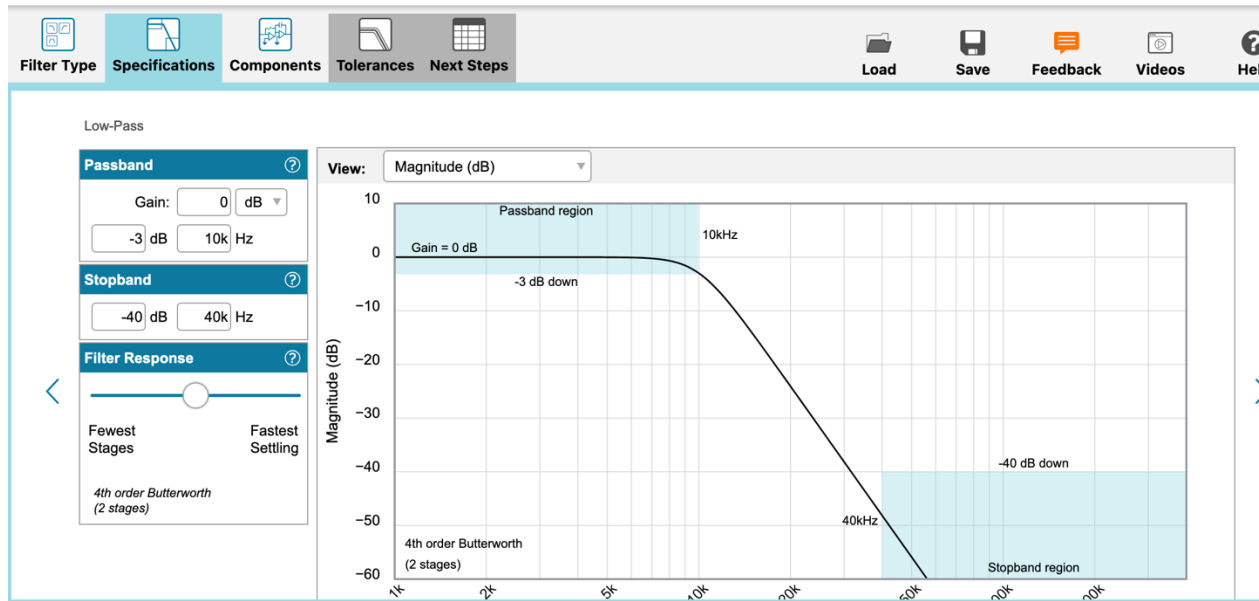


Filter Tools

Analog Devices Filter Tool

(<https://tools.analog.com/en/filterwizard/>)

Analog Filter Wizard



Filter Design Tool

NEW DESIGN LOGIN

Lowpass Filter

FILTER TYPE FILTER RESPONSE TOPOLOGY DESIGN EXPORT

Specification

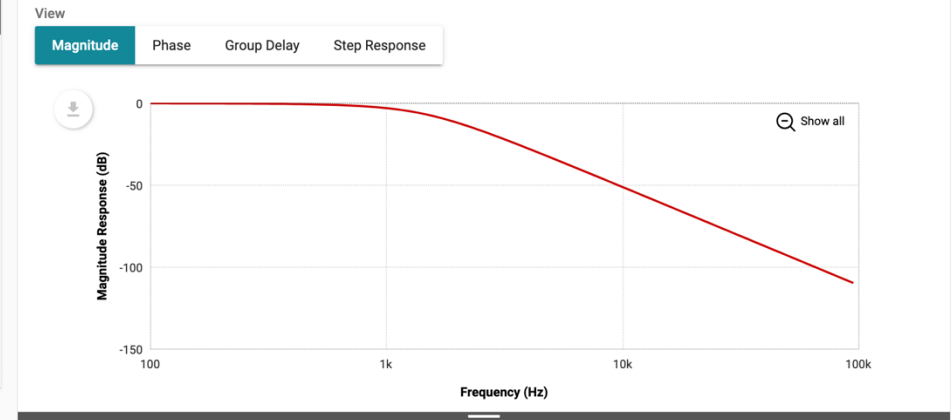
Passband

Gain (Ao) 0 dB V/V

Frequency (Fp) 1000 Hz *

Ripple (Rp) 1 dB

* Cheby's passband is set at the final rolloff of its ripple. All others' passband are set at -3dB



TI Filter Tool

(<https://webench.ti.com/filter-design-tool/>)

Report Criteria

- Discussion on the choice of filter type (i.e., Bessel, Butterworth, Chebyshev)
 - Supported by Python simulation figures
- Analog circuit design methods (use of Filter Tool or Op-amp cook book)
- LT-Spice Circuit Schematic for analog filter
 - Simulated transfer function
- Pictures of breadboard circuit results:
 - Analog filter - Transfer function (use DAD and Waveforms)
 - Input (square wave) and output waveforms for:
 - Analog filter

Resources

- Analog Devices Filter Design Handbook
 - <https://www.analog.com/media/en/training-seminars/design-handbooks/Basic-Linear-Design/Chapter8.pdf>
- Analog Devices Filter Tool (<https://tools.analog.com/en/filterwizard/>)
- TI Filter Tool (<https://webench.ti.com/filter-design-tool/>)

A satellite image of the Earth, showing the continents of Africa and Europe. The image is overlaid with a blue gradient. A bright starburst effect is centered over the continent of Africa.

Filters are Fun!